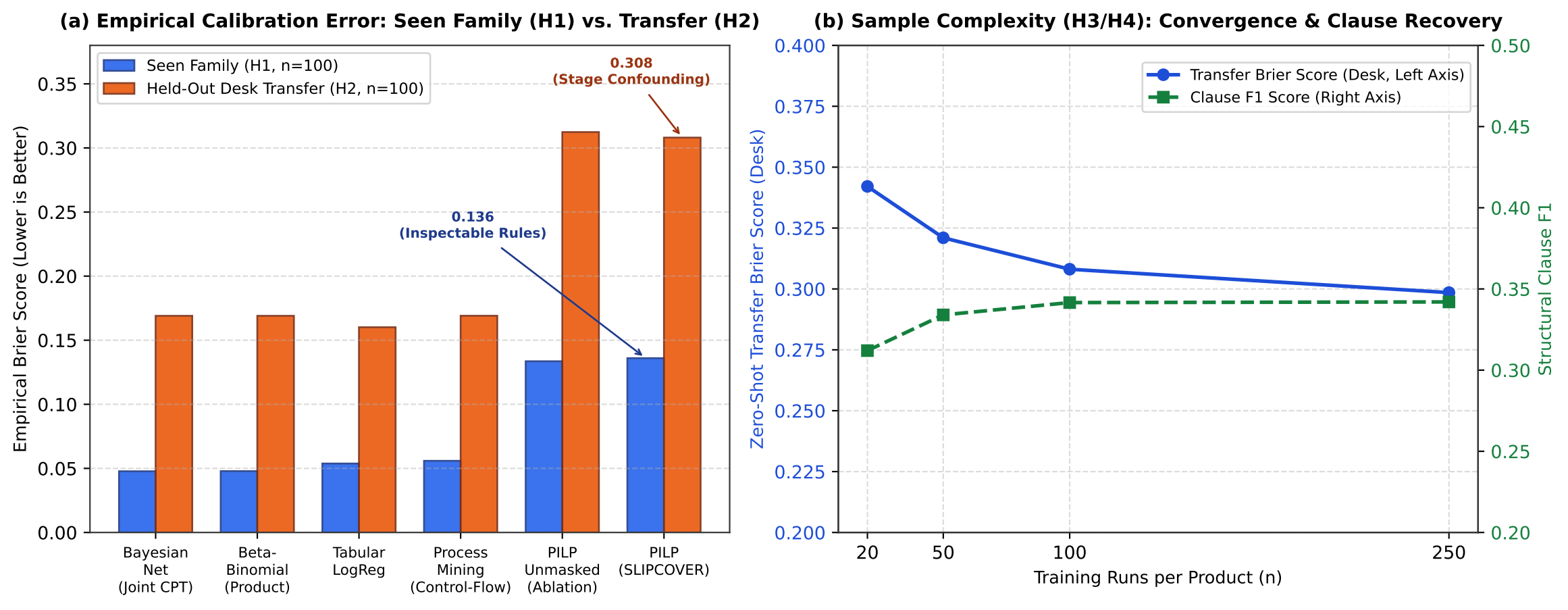
(a) Empirical Calibration Error: Seen Family (H1) vs. Transfer (H2)
Seen Family (H1, n=100)
Held-Out Desk Transfer (H2, n=100)
0.308
(Stage Confounding)
0.136
(Inspectable Rules)
Empirical Brier Score (Lower is Better)
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Bayesian Net (Joint CPT)
Beta-Binomial (Product)
Tabular LogReg
Process Mining (Control-Flow)
PILP Unmasked (Ablation)
PILP (SLIPCOVER)
(b) Sample Complexity (H3/H4): Convergence & Clause Recovery
Transfer Brier Score (Desk, Left Axis)
Clause F1 Score (Right Axis)
Zero-Shot Transfer Brier Score (Desk)
0.200
0.225
0.250
0.275
0.300
0.325
0.350
0.375
0.400
Structural Clause F1
0.20
0.25
0.30
0.35
0.40
0.45
0.50
20
50
100
250
Training Runs per Product (n)